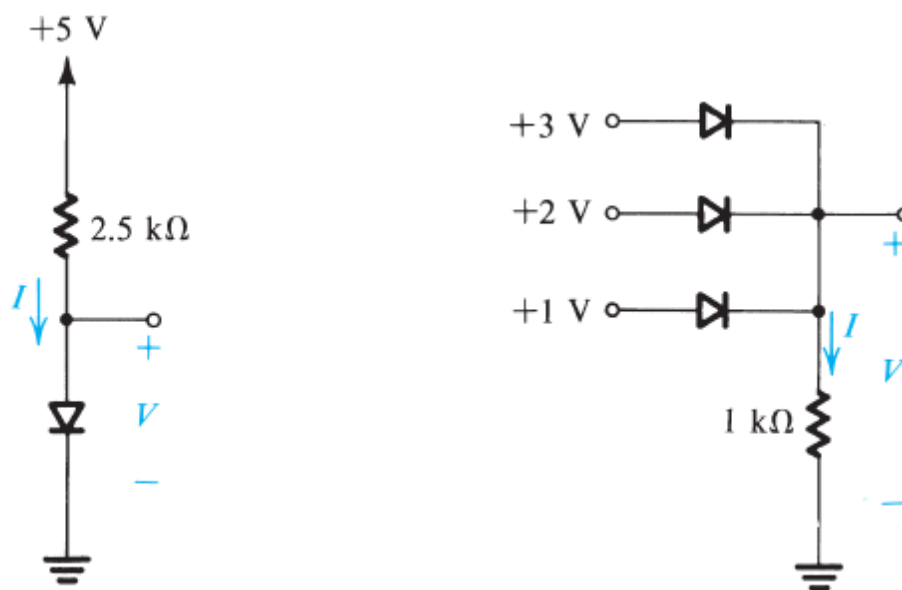
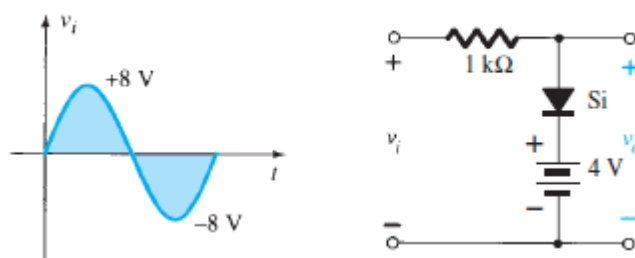


ECI 101 Basics of Electronics Engineering
Home Assignment - 1

1. Find the voltage at which reverse current in Ge diode will reach 90% of its saturation value. Use the diode equation with $\eta = 1$.
2. A Si diode has reverse saturation current of 2.5 micro amp at 300 K. Find forward voltage for a forward current of 10mA. $\eta = 2$ for Si.
3. The current through a Si diode is 60 mA at a forward bias of 0.95 V. Calculate the DC and AC resistance.
4. Find the values of I and V for the given circuits.

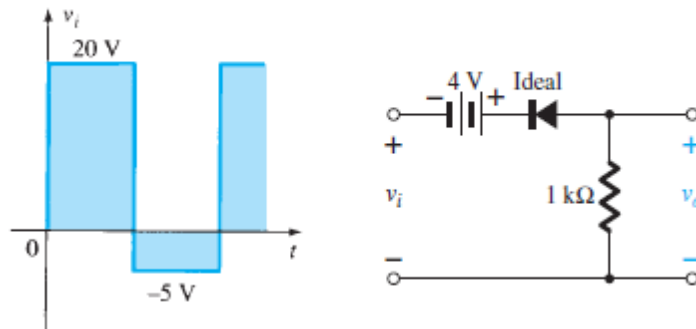


5. A full-wave bridge rectifier with a 120-V rms sinusoidal input has a load resistor of 1 kΩ.
 - a. If silicon diodes are employed, what is the dc voltage available at the load?
 - b. Determine the required PIV rating of each diode.
 - c. Find the maximum current through each diode during conduction.
 - d. What is the required power rating of each diode?
6. Determine v_o for the given circuits
 - (a)



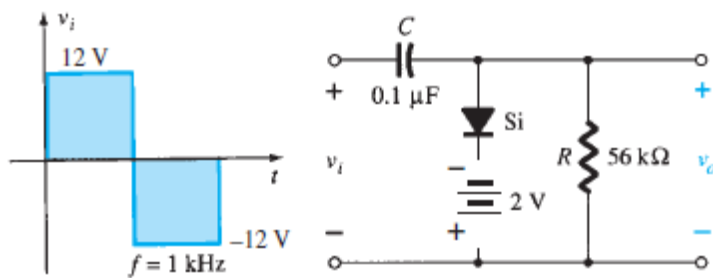
ECI 101 Basics of Electronics Engineering
Home Assignment - 1

(b)



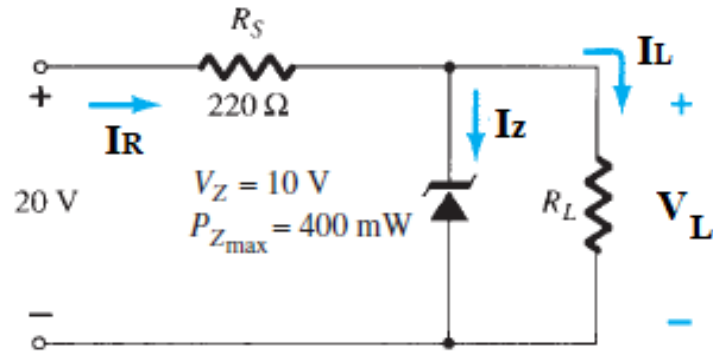
7. For the given network :

- a. Calculate 5τ .
- b. Compare 5τ to half the period of the applied signal.
- c. Sketch v_o .



8.
 - a. Determine V_L , I_L , I_Z , and I_R for the network of Figure shown below if $R_L = 180 \Omega$.
 - b. Repeat part (a) if $R_L = 470 \Omega$.
 - c. Determine the value of R_L that will establish maximum power conditions for the Zener diode.
 - d. Determine the minimum value of R_L to ensure that the Zener diode is in the “on” state.

ECI 101 Basics of Electronics Engineering
Home Assignment - 1



9. A half wave rectifier uses a diode with a forward resistance of 100 ohm. If the input ac voltage is 220V (rms) and the load resistance is of 2 K-ohm. Determine
- Max current, Dc current, rms current,
 - Peak inverse voltage considering the diode is ideal
 - Load output voltage,
 - DC output power and ac output power,
 - ripple factor
 - Rectification Efficiency.
10. The load resistance of a centre-tapped full wave rectifier is 500 ohm and primary to secondary turns ratio is 2:1. The supply to the primary is $240\sin(100\pi t)$ V. Each diode has a resistance of 50 Ohm. Calculate
- peak current, avg current & rms current
 - Ripple factor
 - Efficiency of the rectifier